

Tutorial 10

Linear transformations

1. Which of the following functions are linear? Give reasons for your answers.

(a) $T : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ defined by

$$T(x, y) = (x + y, 2x - y - 3).$$

Solution: Not linear. There are many ways of showing this. A quick way is the following: $T(0, 0) = (0, -3)$. Since $T(0, 0) \neq (0, 0)$, the transformation T is not linear.

Alternatively, show that $T(2, 2) \neq 2T(1, 1)$, for example.

(b) $T : M_{3 \times 3}(\mathbb{R}) \rightarrow \mathbb{R}$ defined by $T(A) = \text{trace}(A)$.

(If you have forgotten what the trace of a square matrix is, look at Question 3 in Tutorial 9.)

Solution: Linear. For any $A, B \in M_{3 \times 3}(\mathbb{R})$, it holds that $\text{trace}(A + B) = \text{trace}(A) + \text{trace}(B)$, and for any $A \in M_{3 \times 3}(\mathbb{R})$ and scalar λ , it holds that $\text{trace}(\lambda A) = \lambda \text{trace}(A)$.

(c) $T : M_{3 \times 3}(\mathbb{R}) \rightarrow \mathbb{R}$ defined by $T(A) = \det(A)$.

Solution: Not linear. There are many counterexamples. For instance, $\det(2I_3) = 8$, while $2 \det(I_3) = 2 \cdot 1 = 2$. Hence $T(2I_3) \neq 2T(I_3)$.

(d) $R : C[0, 1] \rightarrow \mathbb{R}^2$ defined by

$$R(f) = (f(0), \int_0^1 f(x) dx).$$

(Recall that $C[0, 1]$ is the vector space of all real-valued continuous functions defined on the interval $[0, 1]$.)

Solution: Linear. Make use of the properties of integrals to show this.

2. (a) Show that

$$B = \{1 + x, 1 + x^2, x + x^2\}$$

is a basis for $P_2(x)$.

Solution: We know that the dimension of $P_2(x)$ is 3. Hence it is enough to show that the three vectors in B are linearly independent. We solve the equation

$$\lambda_1(1+x) + \lambda_2(1+x^2) + \lambda_3(x+x^2) = 0 + 0x + 0x^2.$$

We collect terms according to equal powers of x :

$$(\lambda_1 + \lambda_2) + (\lambda_1 + \lambda_3)x + (\lambda_2 + \lambda_3)x^2 = 0 + 0x + 0x^2.$$

We get the linear system

$$\lambda_1 + \lambda_2 = 0$$

$$\lambda_1 + \lambda_3 = 0$$

$$\lambda_2 + \lambda_3 = 0.$$

For instance using Gauss elimination, we can see that the only solution of the system is $\lambda_1 = \lambda_2 = \lambda_3 = 0$. Hence the vectors in B are linearly independent, and thus B is a basis for $P_2(x)$.

Alternatively, you can prove that B is a basis for $P_2(x)$ by showing that $\text{span}(B) = P_2(x)$.

(b) It is given that $T : P_2(x) \rightarrow P_1(x)$ is a linear transformation and that

$$T(1+x) = 2-x, \quad T(1+x^2) = 3x, \quad T(x+x^2) = 4+x.$$

Find $T(2-x^2)$.

Solution: We write $2-x^2$ as linear combination of the vectors in the basis B . After some calculation (linear system, Gauss reduction) we get that

$$2-x^2 = 1.5(1+x) + 0.5(1+x^2) - 1.5(x+x^2).$$

Since T is a linear transformation, it follows that

$$T(2-x^2) = 1.5T(1+x) + 0.5T(1+x^2) - 1.5T(x+x^2),$$

and hence by the definition of T ,

$$\begin{aligned} T(2-x^2) &= 1.5(2-x) + 0.5(3x) - 1.5(4+x) \\ &= -3 - 1.5x. \end{aligned}$$

- (c) For $a, b, c \in \mathbb{R}$, find $T(a + bx + cx^2)$.

Solution: After some calculation (linear system, Gauss reduction) we get that for any $a, b, c \in \mathbb{R}$, the vector $a + bx + cx^2$ can be written as linear combination of vectors in B as follows:

$$a + bx + cx^2 = \frac{a - c + b}{2}(1 + x) + \frac{a + c - b}{2}(1 + x^2) + \frac{b - a + c}{2}(x + x^2).$$

Hence

$$T(a + bx + cx^2) = \frac{a - c + b}{2}T(1 + x) + \frac{a + c - b}{2}T(1 + x^2) + \frac{b - a + c}{2}T(x + x^2),$$

and hence by the definition of T ,

$$\begin{aligned} T(a + bx + cx^2) &= \frac{a - c + b}{2}(2 - x) + \frac{a + c - b}{2}(3x) + \frac{b - a + c}{2}(4 + x), \\ &= -a + 3b + c + \frac{a - b + 5c}{2}x. \end{aligned}$$

- (d) Find the range of T .

Solution: Using the solution to (c), it is easy to see that $\text{ran}(T) = P_1(x)$.

- (e) Is $\{T(1 + x), T(1 + x^2), T(x + x^2)\}$ a basis for the range of T ? Give a reason for your answer.

Solution: No. $\{T(1 + x), T(1 + x^2), T(x + x^2)\}$ is not a basis for the range of P_1 . The dimension of $P_1(x)$ is 2, and therefore any basis of $P_1(x)$ must consist of exactly 2 vectors.

3. Let the function $T : M_{2 \times 2}(\mathbb{R}) \rightarrow P_2$ be defined by

$$T\left(\begin{pmatrix} a & b \\ c & d \end{pmatrix}\right) = (a + b)x^2 + (b + c - d)x + (c - d).$$

- (a) A student says: “It is easy to see that T is a linear transformation; it follows from the fact that $T(\mathbf{0}) = \mathbf{0}$.” What is wrong with this argument?

Solution: If a function T is linear, we do have $T(\mathbf{0}) = \mathbf{0}$, but the converse of this statement is **not** true. Therefore we **cannot** conclude from the fact that $T(\mathbf{0}) = \mathbf{0}$ that T is linear. Find an example yourself of a function f between vector spaces for which $f(\mathbf{0}) = \mathbf{0}$ but which is not linear.

- (b) Show that T is linear.

Solution: For any $\lambda \in \mathbb{R}$ and $a, b, c, d \in \mathbb{R}$,

$$\begin{aligned} T\left(\lambda \begin{pmatrix} a & b \\ c & d \end{pmatrix}\right) &= T\left(\begin{pmatrix} \lambda a & \lambda b \\ \lambda c & \lambda d \end{pmatrix}\right) \\ &= (\lambda a + \lambda b)x^2 + (\lambda b + \lambda c - \lambda d)x + (\lambda c - \lambda d) \\ &= \lambda[(a + b)x^2 + (b + c - d)x + (c - d)] \\ &= \lambda T\left(\begin{pmatrix} a & b \\ c & d \end{pmatrix}\right). \end{aligned}$$

In the same way, we can show that for any $a, b, c, d, e, f, g, h \in \mathbb{R}$,

$$T\left(\begin{pmatrix} a & b \\ c & d \end{pmatrix} + \begin{pmatrix} e & f \\ g & h \end{pmatrix}\right) = T\left(\begin{pmatrix} a & b \\ c & d \end{pmatrix}\right) + T\left(\begin{pmatrix} e & f \\ g & h \end{pmatrix}\right).$$

- (c) Recall that the kernel of T is given by $\ker(T) = \{A \in M_{2 \times 2}(\mathbb{R}) : T(A) = \mathbf{0}\}$. Find $\ker(T)$.

Solution: $\ker(T) = \left\{ \begin{pmatrix} 0 & 0 \\ a & a \end{pmatrix} : a \in \mathbb{R} \right\}.$

- (d) Find a basis for the kernel of T . What is $\dim(\ker(T))$?

Solution: A basis for the kernel of T is for instance $\left\{ \begin{pmatrix} 0 & 0 \\ 1 & 1 \end{pmatrix} \right\}$.
It follows that $\dim(\ker(T)) = 1$.

- (e) Is T one-to-one? Give a reason for your answer.

Solution: T is not one-to-one, since $\ker(T) \neq \{\mathbf{0}\}$.

- (f) Recall that the range of T is given by $\text{ran}(T) = \{T(A) : A \in M_{2 \times 2}(\mathbb{R})\}$. Find $\text{ran}(T)$.

Solution: $\text{ran}(T) = P_2$. One way to find the range would be to use Theorem 3.2.15:

Since $\left\{ \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}, \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}, \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix}, \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix} \right\}$ is a basis for $M_{2 \times 2}(\mathbb{R})$, the polynomials

$$T\left(\begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}\right) = x^2, \quad T\left(\begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}\right) = x^2 + x,$$

$$T\left(\begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix}\right) = x + 1, \quad T\left(\begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix}\right) = -x - 1$$

span the range of T .

Hence the range is $\text{span}\{x^2, x^2 + x, x + 1, -x - 1\} = \text{span}\{x^2, x^2 + x, x + 1\}$, and it is relatively easy to see that $\text{span}\{x^2, x^2 + x, x + 1\} = P_2$.

- (g) Find a basis for the range of T . What is $\dim(\text{ran}(T))$.

Solution: A basis for the range of T is $\{x^2, x^2 + x, x + 1\}$. Another possibility would be $\{1, x, x^2\}$. The dimension of the range is 3.

- (h) Is T an onto function (surjective)? Give a reason for your answer.

Solution: Yes, since $\text{ran}(T) = P_2$, by (f).

4. Let V be the vector space of all differentiable functions from $\mathbb{R}$ to $\mathbb{R}$. The function $T : V \rightarrow \mathbb{R}^2$ is defined by:

$$T(f) = (f(0), f'(0)), \quad \text{where } f \in V.$$

- (a) What is $T(f)$ when $f(x) = cx + d$?

Solution: (d, c)

- (b) Is T a linear map? Give reasons for your answer.

Solution: Yes. This follows from the properties of derivatives; make sure that you can write out a proof.

- (c) It is possible to describe the functions in $\ker(T)$ in terms of their graphs. Complete the following description:
“A function is in $\ker(T)$ if and only if its graph ...”

Solution: The kernel of T consists of those differentiable functions from $\mathbb{R}$ to $\mathbb{R}$ whose graphs pass through the origin, and have horizontal tangent lines at the origin.

- (d) Is T one-to-one (injective)? Give reasons for your answer.

Solution: No, since $\ker(T)$ does not consist of the zero function only. For example, the function $f(x) = x^2$ is in the kernel of T .
 (Alternatively, if $f(x) = x^2$ and $g(x) = x^3$, then $f \neq g$, but $T(f) = (0, 0) = T(g)$, so T is not one-to-one.)

- (e) Find the range of T . Is T onto (surjective)?

Solution: $\text{Ran}(T) = \mathbb{R}^2$. To see this note that $\text{ran}(T) \subseteq \mathbb{R}^2$, and that given any arbitrary member (a, b) of $\mathbb{R}^2$, the function h defined by $h(x) = bx + a$ is in V and $T(h) = (a, b)$. Hence T is surjective.

5. Let V and W be vector spaces, $T : V \rightarrow W$ a linear map and $B = \{\mathbf{b}_1, \mathbf{b}_2, \dots, \mathbf{b}_n\}$ be a basis for V . Put $C = \{T(\mathbf{b}_1), T(\mathbf{b}_2), \dots, T(\mathbf{b}_n)\}$.

- (a) Prove that C spans the range of T .

[Hint: If $\mathbf{v} \in \text{ran}(T)$, then there is a $\mathbf{u} \in V$ such that $\mathbf{v} = T(\mathbf{u})$.]

Solution: See Theorem 3.2.15.

- (b) Show that C is not necessarily linearly independent.

[Hint: Give an example of a T and a B for which C is not linearly independent. Question 2 may help you here.]

Solution: See, for example, Example 3.2.16.

- (c) What property must T have to ensure that C will be linearly independent? Give a proof for your answer.

[Hint: Start with $\lambda_1 T(\mathbf{b}_1) + \lambda_2 T(\mathbf{b}_2) + \dots + \lambda_n T(\mathbf{b}_n) = \mathbf{0}$ and then use the fact that T is linear to write the left hand side in a different way.]

Solution: C will be linearly independent if $\{\mathbf{x} \in V : T(\mathbf{x}) = \mathbf{0}\} = \{\mathbf{0}\}$; in other words, T is one-to-one. To see this, note that in this case

$$\begin{aligned} \lambda_1 T(\mathbf{b}_1) + \lambda_2 T(\mathbf{b}_2) + \dots + \lambda_n T(\mathbf{b}_n) = \mathbf{0} &\iff T(\lambda_1 \mathbf{b}_1 + \lambda_2 \mathbf{b}_2 + \dots + \lambda_n \mathbf{b}_n) = \mathbf{0} \\ &\iff \lambda_1 \mathbf{b}_1 + \lambda_2 \mathbf{b}_2 + \dots + \lambda_n \mathbf{b}_n = \mathbf{0} \\ &\iff \lambda_1 = \lambda_2 = \dots = \lambda_n = 0. \end{aligned}$$

As a challenge, prove that the converse is also true.

- (d) What property must T have to ensure that C will be a basis for $\text{ran}(T)$?

Solution: $\{\mathbf{x} \in V : T(\mathbf{x}) = \mathbf{0}\} = \{\mathbf{0}\}$, or equivalently T is one-to-one.

6. In this question you have to use your knowledge of bases and coordinate vectors to prove something about a function between two vector spaces.

Let V be an n -dimensional real vector space and $B = \{\mathbf{b}_1, \mathbf{b}_2, \dots, \mathbf{b}_n\}$ an ordered basis for V . If $\mathbf{v} \in V$, we write $[\mathbf{v}]_B$ for the coordinate vector of $\mathbf{v}$ with respect to the basis B .

- (a) Prove that if $\mathbf{u}, \mathbf{v} \in V, \lambda \in \mathbb{R}$, then $[\lambda\mathbf{u}]_B = \lambda[\mathbf{u}]_B$ and $[\mathbf{u} + \mathbf{v}]_B = [\mathbf{u}]_B + [\mathbf{v}]_B$.
[Hint: There is a unique way of writing $\mathbf{u}$ as a linear combination of elements of B ; the scalars used give you $[\mathbf{u}]_B$. To find $[\lambda\mathbf{u}]_B$, you need to express $\lambda\mathbf{u}$ as a linear combination of elements of B ; there is only one way of doing it. Find it! The same procedure works for $[\mathbf{u} + \mathbf{v}]_B$.]

Solution: Let $B = \{\mathbf{b}_1, \mathbf{b}_2, \dots, \mathbf{b}_n\}$ be the ordered basis for V , $\mathbf{u}, \mathbf{v} \in V$ and $\lambda \in \mathbb{R}$. Then each of $\mathbf{u}, \mathbf{v}$ can be written in a unique way as a linear combination of elements of B :

$$\mathbf{u} = \lambda_1 \mathbf{b}_1 + \lambda_2 \mathbf{b}_2 + \dots + \lambda_n \mathbf{b}_n$$

$$\mathbf{v} = \mu_1 \mathbf{b}_1 + \mu_2 \mathbf{b}_2 + \dots + \mu_n \mathbf{b}_n.$$

By definition of coordinate vectors, $[\mathbf{u}]_B = (\lambda_1, \lambda_2, \dots, \lambda_n)$ and $[\mathbf{v}]_B = (\mu_1, \mu_2, \dots, \mu_n)$. Since

$$\lambda\mathbf{u} = \lambda\lambda_1 \mathbf{b}_1 + \lambda\lambda_2 \mathbf{b}_2 + \dots + \lambda\lambda_n \mathbf{b}_n \text{ and}$$

$$\mathbf{u} + \mathbf{v} = (\lambda_1 + \mu_1) \mathbf{b}_1 + (\lambda_2 + \mu_2) \mathbf{b}_2 + \dots + (\lambda_n + \mu_n) \mathbf{b}_n,$$

we have, again by definition of coordinate vectors,

$$\begin{aligned} [\lambda\mathbf{u}]_B &= (\lambda\lambda_1, \lambda\lambda_2, \dots, \lambda\lambda_n) = \lambda(\lambda_1, \lambda_2, \dots, \lambda_n) = \lambda[\mathbf{u}]_B \\ [\mathbf{u} + \mathbf{v}]_B &= (\lambda_1 + \mu_1, \lambda_2 + \mu_2, \dots, \lambda_n + \mu_n) \\ &= (\lambda_1, \lambda_2, \dots, \lambda_n) + (\mu_1, \mu_2, \dots, \mu_n) \\ &= [\mathbf{u}]_B + [\mathbf{v}]_B. \end{aligned}$$

- (b) Let the function $F : V \rightarrow \mathbb{R}^n$ be defined by $F(\mathbf{v}) = [\mathbf{v}]_B$. What does (a) tell you about the function F ?

Solution: F is a linear transformation from V to $\mathbb{R}^n$.

- (c) Is the function F defined in (b) one-to-one? Give a reason for your answer.

Solution: Yes, F is one-to-one:

Suppose that $F(\mathbf{v}) = F(\mathbf{w})$. Then by definition, $[\mathbf{v}]_B = [\mathbf{w}]_B$, which means that the coordinate vectors of $\mathbf{v}$ and $\mathbf{w}$ relative to the basis B are equal. Hence $\mathbf{v} = \mathbf{w}$.

- (d) Is $\{[\mathbf{b}_1]_B, [\mathbf{b}_2]_B, \dots, [\mathbf{b}_n]_B\}$ a basis for $\mathbb{R}^n$? Give a reason for your answer.

Solution: Yes. The given set is just the standard basis for $\mathbb{R}^n$.